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Volatility dynamics of crypto-currencies' returns: Evidence from asymmetric and long memory GARCH models

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ABSTRACT

The objective of this paper is to select the most optimum model or set of models useful for modeling sixteen of the most popular crypto-currencies associated volatility. Five GARCH models, with different error distributions, are fitted to each of these crypto-currencies. The most effectively fit model or superior set of models is then selected through maximizing the likelihood and minimizing the AIC and BIC information criteria. The reached results prove that the majority of crypto-currencies turn out to be rather effectively modulated via the TGARCH with double exponential distribution. Indeed, the attained findings report an asymmetric effect whereby volatility turns out to increase rather by response to positive shocks than by response to negative shocks, implying an asymmetric effect that differs from that generally observed in stock markets. The increase in volatility, as emanating in response to positive shocks may well have its justification in the uninformed investors' undertaken herding strategies.

1. Introduction

The recent decade has been noticeably marked with a spectacular development of crypto-currencies. Initially devised by Nakamoto (2008), the Bitcoin stands as the first decentralized crypto-currency, primarily based on the blockchain technology. The Bitcoin has helped remarkably in facilitating electronic payments among individuals without the need to go through the intermediacy of a third party. It is the first devised digital currency, and still remains the market leading crypto-currency in terms of stock market capitalization. For the period ranging from November 2016 to November 2017, the Bitcoin's market capitalization scored an increase from 10 to 79 billion Dollars. It is actually this significant growth which has induced and enhanced the launch of other crypto-currencies. Today, there exist more than1700 crypto-currencies; Crypto-currencies are a growing asset class, with a total market capitalization of US$ 133 billion by December 31, 2018. The peak of this capitalization value is reached on January 6, 2018 with a ceiling of US$ 800 billion.

In this paper, the focus of interest is laid on studying 16 crypto-currencies standing as the major crypto-currencies with respect to market capitalization and volume. It is important to note that the Bitcoin started to be traded in 2010, on the Mt. Gox stock exchange. As for Augur and Ether, they are crypto-currencies whose block-chains are generated by the Ethereum platform. Another currency, the BitShares, devised by Daniel Larimer, has been described by the crypto-currency platform as a digital currency in 2013. Dash is an open source crypto-currency, an Altcoin that was forked from the Bitcoin protocol. EOS.IO is a block-chain protocol powered by the native crypto-currency EOS. Founded by David Sonstebo, the IOTA is a crypto-currency focused on providing payments and secure communications between machines on the Internet of Things. Komodo is a decentralized ICO platform and a Zero knowledge crypto-

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currency. Lisk is a block-chain application platform. It is based on its own Block-chain network and token LSK. Created in April 2014, Monero is an open-source crypto-currency focused on decentralization, fungibility, andprivacy.NEO is a block-chain platform and crypto-currency designed to construct a scalable network of decentralized applications. Qtum is a Chinese crypto-currency and hybrid platform. It connects the existing blockchain with virtual machines. Ripple is instead based on the Ripple protocol, which is a currency exchange and remittance network, initially released in 2012. Stellar is an open-source decentralized protocol for digital currency to “fiat” money transfers, enabling cross-border transactions to be maintained between any pair of currencies. Stratis is a flexible block-chain that stands as a powerful development platform designed to satisfy the businesses’ needs in real-world financial services. The Waves Platform is a crypto-currency project which designates the analogue of the Kickstarter crowd funding platform.

In this context, the present study is designed to select the best model or set of models under different error distribution useful for effectively modeling volatility relating to sixteen most popular crypto-currencies. Noteworthy, in this respect, is that several studies have been conducted to investigate the stock indices volatility as associated to commodities and crypto-currencies by exclusively focusing on the shock-asymmetry dimension (EGARCH model, Dyhrberg, 2016a; Dorfleitner and Lung, 2018), or through the long-memory aspect (ARIMA model, Bouri et al., 2018). In effect, neglecting to simultaneously account for these stylized facts as a single set might well result in a biased estimation of volatility, and, there from, a biased investor decision in regard to portfolio diversification.

As a matter of fact, both of the long-memory properties and asymmetric effects represent remarkable stylized factors, likely to bring about significant financial outcomes. In general, a time series is considered to exhibit a long-memory and asymmetric behavior once the relating autocorrelation function proves to be non-integrable. Actually, the major distinctive advantage associated with the long-memory dimension lies in the nonlinear dependence it reflects throughout the first and second moments (Elder and Serletis, 2008). Additionally, a high long-memory related coefficient helps well reflect whether a certain asset displays long positive or negative strays from the equilibrium, there from, whether such a commodity is no longer representing an effective hedging or a safe haven instrument, due to the failure it exhibits in providing an effective shield against adverse market conditions.

Noteworthy, however, is that, like most of the previously conducted study cases dealing with the Bitcoin price volatility issue while applying a single conditional heteroskedasticity model, a major persistent question remains still unanswered as to which conditional heteroskedasticity model proves to be conveniently fit for effectively explaining the Bitcoin relevant data. Worth citing, in this respect, are the studies conducted by Bouoiyour and Selmi (2016); Katsiampa (2017) as well as Charle and Darne-Lemna (2018) that consider comparing some of the GARCH-type models without investigating both of the long-memory and asymmetric-effect relevant properties associated with the Bitcoin. Still, Baur and Dimpfl (2018); Charfeddine and Maouchi (2018); Peng et al. (2018) and Caporale and Zekokh (2019) have made attempts to select the best appropriate model or superior set of GARCH volatility models fit for examining a number of crypto-currencies, but without mentioning the properties relevant to both of the long-memory and asymmetric-effect factors with different error distribution. In this context, given the remarkably rapid surge of crypto-currencies as new and emerging, the paper’s contribution lies in investigating the most convenient framework among the two asymmetric and long-memory conditional heteroskedasticity models with different error distribution, such as Student, normal, generalized error distribution and double exponential distribution, proves to fit most for effectively describing the sixteen crypto-currencies associated volatilities.

The remainder of this paper is organized as follows. Section 2 is devoted to provide a general review of related literature. Section 3 serves to discuss our applied econometric methodology. Section 4 is dedicated to highlighting the relevant data and reached empirical findings. Finally, Section 5 is reserved to highlight the major relevant conclusions.

2. Literature review

Cheah and Fry (2015); Corbet et al. (2017) and Hafner (2018) document that the Bitcoin proves to stand at the origin of extreme volatilities and bubbles, owing mainly to the speculative purposes it implicitly exhibits. Additionally, the Bitcoin based transactions are usually anonymous and concluded at low costs (Kim, 2017), while providing some diversification benefits (Corbet et al., 2018). In its full form, the Bitcoin is mainly regarded as an asset rather than a currency (Glaser et al., 2014; Baek and Elbeck, 2015; Dyhrberg, 2016a). In this respect, several recently conducted studies have been focused on investigating the Bitcoin returns associated volatility dynamics, using various GARCH-type models. For instance, Glaser et al. (2014) and Gronwald (2014) have applied the linear GARCH, while Dyhrberg (2016b); Bouoiyour and Selmi (2015, 2016) along with Bouri et al.(2017) have appealed to the Threshold GARCH (TGARCH) model. On implementing asymmetric based GARCH models, Bouri et al. (2017); Katsiampa (2017); Baur et al. (2018) and Stavroyiannis (2018) proposed to examine the conditional variance response to past-positive and-negative shocks to discover the persistence of an inverted leverage effect. In turn, and on estimating the Bitcoin relating volatility through comparing a selection of GARCH models, Katsiampa (2017) reached the conclusion that the AR-CGARCH model proves to display the most optimal fit level. As for Charle and Darne-Lemna (2018), they have replicated Katsiampa (2017) conducted study and achieved partially different results. Using the robust QML estimator to compute the parameters’ standard errors, the authors deduced that the six GARCH-type models, subject of their study, proved to be inappropriate for modeling the Bitcoin returns.

Most of the relevant works seem to analyze returns and treat volatility associated with the Bitcoin as a leading market crypto-currency. A number of new crypto-currencies have made their appearance, most of which are further developed on the block-chain basis. With respect to Baur and Dimpfl (2018), they considered to analyze the asymmetric volatility effects as related to 20 different crypto-currencies. They reported that positive shocks prove to increase volatility more than negative shocks do, implying an asymmetric effect that is different to that generally observed in stock markets. They allotted these financial assets relating a typical effect to the informed traders’ undertaken trading activity concerning negative shocks, and to the uninformed investors’ herding and buying trading activity regarding positive shocks. As for Charfeddine and Maouchi (2018), they investigate the Bitcoin, Ethereum, Ripple, and LiteCoin respective volatility series' EMH and LRD behaviors. They found that the long memory characterizing the returns and volatilities of Bitcoin, LiteCoin and Ripple did not prove to emanate from "methodological illusion".

Based on eleven cryptocurrencies, Kaiser (2018) proposed to test the existence of well-known equity seasonality patterns for crypto-currency returns' daily volatility. They did not notice the persistence of remarkably consistent and robust calendar effects in crypto-currency returns. Mensi et al. (2018) explored the impact of dual long-memory and structural breaks on the conditional volatility of the Bitcoin and Ethereum markets. On implementing four different models, more specifically, ARFIMAGARCH, ARFIMA-FIGARCH, ARFIMA-FIAPARCH, and ARFIMA-HYGARCH, they reached the conclusion that the dual long-memory property of the Bitcoin and Ethereum proved to refute the market efficiency and random walk hypothesis. Their attained findings revealed that the mean and variance equations' long-memory aspect turns out to decrease significantly. They also outlined that the FIGARCH model with SB variables appeared to provide a comparatively superior forecasting capacity as compared to the other models. Peng et al. (2018) evaluated the predictive performance of three crypto-currencies relevant volatility (the Bitcoin, ethereum and dash) and that associated to three currencies (the Euro, GB Pound and Japanese Yen (in US dollars)) with recognized value stores, using daily and hourly frequency data. They jointly combined the traditional GARCH model and the Machine Learning framework to estimate the relevant mean and volatility equations, using Support Vector Regression. They came to conclude that the SVR-GARCH models proved to outperform the GARCH, EGARCH and GJR-GARCH models with Normal, and Skewed Student's t distributions.

On studying 224 Crypto-currencies, Phillip et al. (2018a) demonstrated that crypto-currencies proved to exhibit noticeable long-memory, heavy tailedness, leverage as well as stochastic volatility dimensions. Phillip et al. (2018b) applied a Gegenbauer long memory stochastic volatility model with leverage and a bivariate Student's t-error distribution to capture the major stylized facts related to114 crypto-currencies. The results proved that the Ripple appeared to exhibit the lowest over-night risk compared to the entirety of the other crypto-currencies. In turn, Tan et al. (2018) suggested to measure five of the crypto-currencies related volatilities using the Garman and Klass volatility measures. They modeled the volatility measures through asymmetric bilinear Conditional Autoregressive Range (ABL-CARR) model. Their achieved findings showed that the Bitcoin Cash appeared to demonstrate greater volatility rates, therefore, standing as noticeably riskier than the Bitcoin, Ethereum, Litecoin and Ripple. The Bitcoin Cash behaves more like a speculative asset than a digital currency. It represents a convenient choice for speculative investors. As for the Bitcoin and Litecoin, they demonstrated higher kurtosis, stronger persistence and lower leverage effect and volatility features, implying that they display a rather remarkable predictable volatility, lower volatility ranges, and are, therefore, less risky. Concerning the Ethereum, it displayed such volatility features as moderate persistence and auto-regression as well as a mild leverage effect. Regarding the Ripple, however, it proved to exhibit high auto-regression scores, but a leverage and mild persistence approaching zero.

Using the spillover-index approach, Yi et al. (2018) studied both static and dynamic volatility connectedness among 8 crypto-currencies. They also used the LASSO to shrink and estimate VAR parameters to construct volatility connectedness network. They discovered that the total volatility connectedness among eight crypto-currencies proved to fluctuate periodically, increasing when the market is experiencing unpredictable exogenous shocks or unstable economic conditions. Crypto-currencies with high market capitalization properties, like the Bitcoin, Dogecoin and Litecoin propagate large volatility shocks, while small market capitalization crypto-currencies are more likely to receive volatility shocks from others. By sing the VaR and ES backtesting as well as the MCS procedure, Caporale and Zekokh (2019) envisaged to select the best model or superior set of GARCH volatility models fit for Bitcoin, Ethereum, Ripple and Litecoin. Their reached results proved to highlight that two-regime GARCH models appeared to yield better VaR and ES predictions than did the single-regime models. The standard GARCH model was suggested as appropriate for the Bitcoin's case, while the GJR GARCH specification model was rated as suitable for treating the Ethereum's case. As for the Ripple and Litecoin, the authors were led to opt for the standard GARCH. Using the COVAR, Borri (2019) estimated the conditional tail-risk in the markets for Bitcoin, Ether, Ripple and Litecoin. He concluded that these crypto-currencies are highly exposed to tail-risk within the crypto-market contexts. They also discovered that crypto-currency returns are highly correlated with one another.

To sum it up, the Bitcoin related volatility properties have recently been thoroughly analyzed, while the other crypto-currencies associated volatility has received less attention, especially through introducing both of the asymmetry and long memory related effects within a single model, specifically, the FIEGARCH model, and under different error distributions.

3. Econometric methodology

In this section, we consider estimating the different GARCH models under different error-distribution conditions, such as normal, student, generalized-error and double-exponential distributions.

According to the basic GARCH model, since only squared residuals ε_{t-i}^2 can be introduced in the equation, the residuals or shocks associated signs prove to have no effects on conditional volatility. Within a stylized fact of financial volatility, however, bad news (negative shocks) tends to have a more significant impact on volatility than good news (positive shocks). Black (1976) attributed this effect to the fact that bad news tend to drive down the stock price, thus increasing the stock related leverage (i.e., the debt-equity ratio), resulting in a higher stock volatility level. Based on this conjecture, the asymmetric news' impact is usually referred to as the leverage effect. This subsection focuses on treating this issue from the EGARCH, TGARCH and PGARCH models' perspective.

4. EGARCH model

Nelson (1991) proposed the following exponential GARCH (EGARCH) model to account for leverage effects:

$$h_t = a_0 + \sum_{i=1}^{p} a_i \frac{|\varepsilon_{t-i}| + \gamma_i \varepsilon_{t-i}}{\sigma_{t-i}} + \sum_{j=1}^{q} b_j h_{t-j} \tag{1}$$

Where: $h_t = log\sigma_t^2$ or $\sigma_t^2 = e^{ht}$. Note that when ε_{t-i} is positive, or "good news" prevail, the total effect of ε_{t-i} is $(1 + \gamma_i)|\varepsilon_{t-i}|$; inversely, when ε_{t-i} is negative, or there is prevalence of "bad news", the total effect of ε_{t-i} is $(1 - \gamma_i)|\varepsilon_{t-i}|$.Thus, bad news can have a more noticeable impact on volatility, and the value of γ_i wouldbe expected to be negative.

5. TGARCH model

Another GARCH variant, capable of modeling leverage effects, is the threshold GARCH (TGARCH) model, written as:

$$\sigma_t^2 = a_0 + \sum_{i=1}^{P} a_i \varepsilon_{t-i}^2 + \sum_{i=1}^{P} \gamma_i S_{t-i} \varepsilon_{t-i}^2 + \sum_{j=1}^{q} b_j \sigma_{t-j}^2 \tag{2}$$

Where: $S_{t-i} = \begin{cases} 1\, if\ \varepsilon_{t-i} < 0 \\ 0\, if\ \varepsilon_{t-i} \geq 0 \end{cases}$

Accordingly, depending on whether ε_{t-i} is above or below the zero threshold value, ε_{t-i}^2 turns out to have different effects on the conditional variance σ_t^2, i.e., when ε_{t-i} is positive, the total effects are given by $a_i \varepsilon_{t-i}^2$, and when ε_{t-i} is negative,the total effects are given by $(a_i + \gamma_i)\varepsilon_{t-i}^2$. Hence, one would expect γ_i to bepositive for bad news to have more noticeable impacts. This model is also known as the GJR model, because Glosten et al. (1993) proposed essentially the same model.

6. PGARCH model

The basic GARCH model has also been extended to allow for leverage effects to be considered. This is made possible by treating the basic GARCH model as a special case of the power GARCH (PGARCH) model, as initially proposed by Ding, Granger and Engle (1993):

$$\sigma_t^d = a_0 + \sum_{i=1}^{P} a_i (|\varepsilon_{t-i}| + \gamma_i \varepsilon_{t-i})^d + \sum_{j=1}^{q} b_j \sigma_{t-j}^d \tag{3}$$

Where: d is a positive exponent, and γi denotes the leverage effects related coefficient. Note that when d = 2, the Eq. (3) proves to reduce to the basic GARCH model with leverage effects. Ding et al. (1993) outlined that the PGARCH model may also include several other GARCH variants as special cases.

For the autocorrelation problem, stylized facts and shock asymmetry issues to be eliminated among financial series, appeal should be made to the FIGARCH and the FIEGARCH models.

7. FIGARCH model

The basic GARCH (1, 1) model may take the form of an ARMA(1, 1) model in terms of squared residuals. In the same spirit of the GARCH(p, q) model, it can be written as:

$$\sigma_t^2 = a + \sum_{i=1}^{P} a_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} b_j \sigma_{t-j}^2 \tag{4}$$

showing clearly that it can be more easily rewritten as follows:

$$\phi(L)\varepsilon_t^2 = a + b(L)u_t \tag{5}$$

Where: $u_t = \varepsilon_t^2 - \sigma_t^2$

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_m L^m$$

$$b(L) = 1 - b_1 L - b_2 L^2 - \cdots - b_q L^q$$

With: $m = max(p, q)$ and $\phi_i = a_i + b_i$. Obviously, Eq. (5) represents an $ARMA\,(m, q)$ process in terms of squared residuals ε_t^2,with u_t designating an MDS disturbance term.

The predominance of a high persistence in GARCH models suggests that the polynomial $\phi(z) = 0$ may well have a unit root, case in which the GARCH model turns out to be the integrated GARCH (IGARCH) model. To allow for high persistence and long memory to prevail within the conditional variance, while avoiding the sophistications marking the IGARCH models, we consider extending the $ARMA(m, q)$ process, as appearing in (5),into a $FARIMA(m, d, q)$ process as follows:

$$\phi(L)(1 - L)^d \varepsilon_t^2 = a + b(L)u_t \tag{6}$$

Where the entirety of the roots of $\phi(z) = 0$ and $b(z) = 0$ lie outside the unit circle. Once d = 0, the usual GARCH model turns out to be reduced; when d = 1, the turns out to be transformed into an IGARCH model; when 0 < d < 1, the fractionally differenced squared residuals, $(1-L)^d\varepsilon_t^2$, follow a stationary ARMA(m, q) process. Hence, the above FARIMA process relevant to ε_t^2 can be rewritten in terms of the conditional variance σ_t^2 as:

$$b(L)\sigma_t^2 = a + [b(L) - \phi(L)(1-L)^d]\varepsilon_t^2 \tag{7}$$

Baillie et al., 1996 dubbed the above-figuring model as the fractionally integrated GARCH, or FIGARCH(m, d, q) model. When$0 < d < 1$, the coefficients associated with $\phi(L)$ and $b(L)$ prove to capture the short-run dynamics of volatility, while the fractional difference parameter d models help capture the long-run characteristics of volatility.

8. FIEGARCH model

The FIGARCH model helps directly extend the ARMA representation of squared residuals, resultant from the GARCH model, into a fractionally integrated model. However, to ensure that a general FIGARCH model is stationary and that the conditional variance σ_t^2 is always positive, some intractable, often sophisticate; restrictions seem imposed to implement on the model coefficients (see, for instance, Baillie et al. (1996) or Bollerslev and Mikkelsen (1996) for a more thorough discussion).

It is worth noting that an EGARCH model can be represented under an ARMA form, in terms of the conditional variance logarithm, thus, maintainingthat the conditional variance is persistently positive. In this respect, Bollerslev and Mikkelsen (1996) put forward the following fractionally integrated EGARCH (FIEGARCH)model:

$$\phi(L)(1-L)^d \ln\sigma_t^2 = a + \sum_{j=1}^{q}(b_j|x_{t-j}| + \gamma_j x_{t-j}) \tag{8}$$

Where: $\phi(L)$ stands as already defined in the FIGARCH model's case, $\gamma_j \neq 0$ to allow for the maintaining of leverage effects, and x_t designates the standardized residual, such as:

$$x_t = \frac{\varepsilon_t}{\sigma_t} \tag{9}$$

Besides, Bollerslev and Mikkelsen (1996) highlighted that the FIEGARCH model is stationary if $0 < d < 1$.

9. Data and results

9.1. Data and descriptive statistics

Adjusted closing-price data relevant to sixteen crypto-currencies are applied in this study. The database was collected from the CoinMarketCap and ABC bourse, regarding the period ranging from August,07, 2017 to December, 12, 2018, on a daily frequency basis, making up a total of 366 observations. Daily returns are defined by $r_t = \ln(p_t/p_{t-1})$, with p_t standing for the crypto-currency respective closing price on day t. Fig. 1, below, illustrates the return evolution over time.

Descriptive statistics, relevant to the sixteen observed crypto-currencies, are depicted on the following table (Table 1):

All results are achieved by means of the S-Plus 6 finmetrics 1.0 software module. Based on the above figuring table, one could well notice a decrease marking the entirety of the crypto-currencies' average return. Additionally, estimated risk, as depicted through the standard deviation, is relatively high for the IOTA, NEO, QTUM, Stellar and Stratis crypto-currencies associated returns. As for the Skewness values, they reveal well that most of the marginal distributions turn out to be skewed to the right (with positive related values), while the remaining marginal distributions are discovered to be skewed to the left (displaying negative values). This result may have its explanation in the predominance of rare events. The kurtosis values, relevant to our sample, prove to be greater than the value 3. The kurtosis high associated values indicate well that these cryptocurrencies prove to exhibit flattened distributions with extreme values on the tails. The higher values associated with the Jarque Bera test indicate well that these distributions are abnormal.

Concerning the Lagrange multiplier test, it outlines the prevalence of an ARCH effect in all the return series, except for the Bitcoin, DASH, EOS, Ethereum, LISK and Stellar crypto-currencies, regarding which the relating LM statistic proves to be non-significant. Consequently, these crypto-currencies should be eliminated from our sample, as weare working with models involving an ARCH effect. As a matter of fact, the modified R/S test pertaining results attest well the persistence of long memory with regard to the entirety of the return series.

Table 2, below, illustrates the various GARCH-type models related estimation results. As can be noticed, the AIC and BIC information criteria associated values are discovered to be minimized, and the log likelihood (LL) value maximized under the AR(1)-TGARCH(1,1,1) model with a double exponential distribution with regard to the Augur, Bitshares, Monero, NEO, Ripple and Waves related return-volatility series. Indeed, the three information criteria (AIC, BIC, LL) are also best fit to be opted for by means of the AR (1)-TGARCH(1,1,1) model under generalized error distribution, concerning the Komodo and Stratis relevant return series, while the IOTA returns are more appropriately fit to be modulated through an AR(1)-EGARCH(1,1) under double exponential distribution, and, finally, the AR(1)-EGARCH(1,1) under student distribution proves to stand as the best appropriate tool fit to describe the QTUM return related volatility.

As regards the ARCH(12) and Q tests reached results, which have been used as diagnostic tests, respectively applied to assess the squared and squared standardized residuals relevant to the different appropriate models, they indicate that this very model proves to stand as remarkably suitable for describing the volatility returns, since the hypotheses of no remaining ARCH effects and no

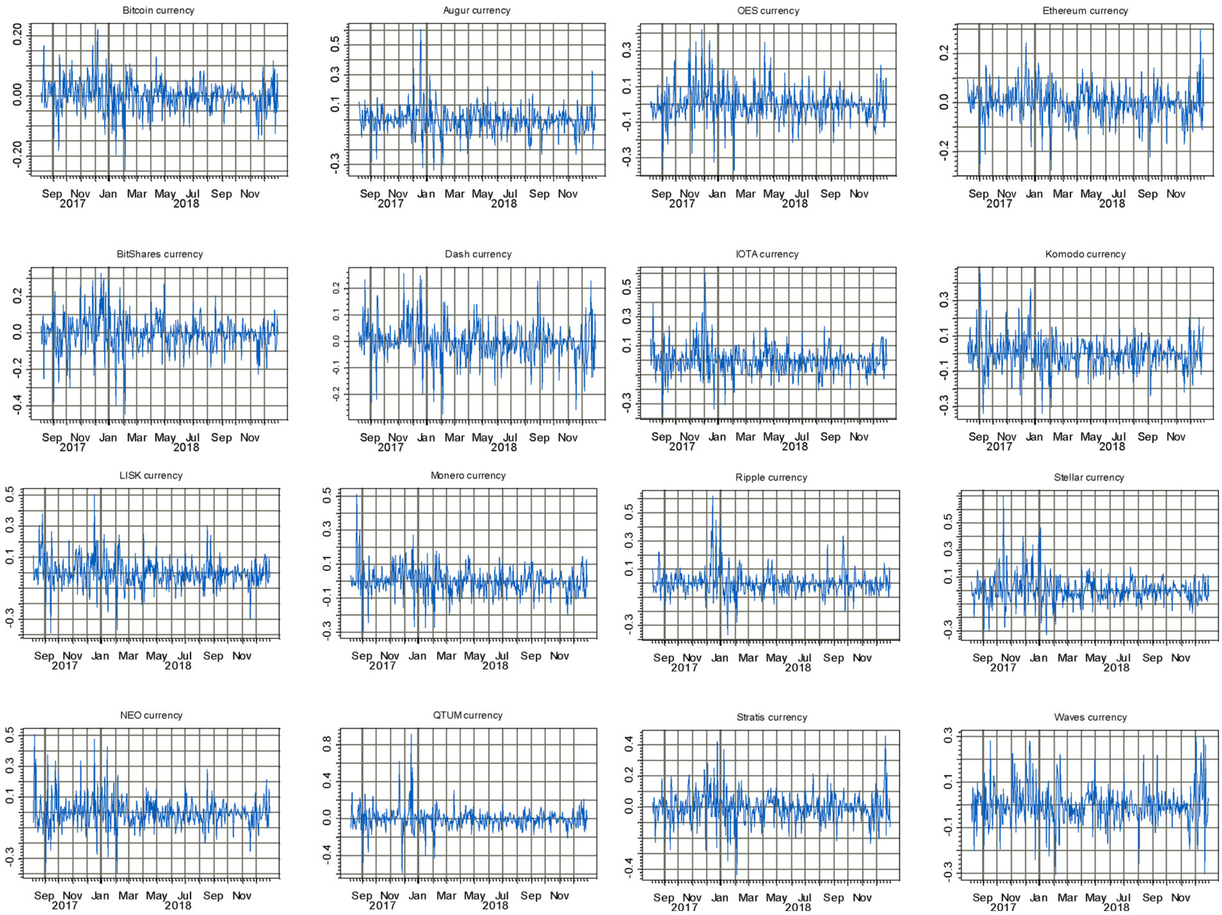

Fig. 1. Price returns recorded evolution.

Table 1
Descriptive statistics.

	Mean	Std Dev	Skew	Kurt	J.B	LM	Q(12)	$Q^2(12)$	R/S mod
Bitcoin	0.00024	0.0551	−0.205	5.406	90.632***	**27.42**	23.58*	33.39***	2.402***
Augur	−0.00230	0.0955	0.735	9.551	685.77***	87.21***	18.74	71.41***	2.504***
Bitshares	−0.00377	0.0998	−0.316	5.595	108.52***	62.66***	32.75***	52.94***	2.450***
DASH	−0.00236	0.0734	0.079	5.384	86.873***	**28.15**	17.95	21.40	1.536
EOS	0.00102	0.0993	0.408	6.003	147.34***	**24.55**	20.19	23.56*	1.958**
Ethereum	−0.00184	0.0695	−0.097	5.339	83.804***	**22.34**	14.14	13.93	1.266
IOTA	−0.00074	0.1027	0.795	7.468	342.24***	64.49***	23.91*	72.57***	1.768
Komodo	−0.00030	0.0962	0.375	5.778	126.01***	44.85***	18.96	68.59**	2.186***
LISK	−0.00110	0.0988	0.514	6.438	195.88***	**25.03**	18.53	27.75**	1.975**
Monero	−0,00007	0.0825	0.471	8.232	429.90***	55.69***	18.99	38.34***	1.812
NEO	−0.00238	0.1053	0.916	7.491	357.89***	61.05***	16.61	36.97***	2.077**
QTUM	−0.00472	0.1183	1.345	17.96	3514.2***	70.80***	30.93**	60.94***	1.923**
Ripple	0.00198	0.0913	1.621	12.46	1521.8***	76.60***	22.69	92.60***	1.913**
Stellar	0.00452	0.1070	1.282	9.821	807.56***	**31.94**	16.33	36.69***	2.31***
Stratis	−0.00510	0.1032	0.230	5.887	130.06***	66.31***	25.97*	57.87***	2.048**
Waves	−0.00160	0.0882	0.368	5.194	81.488***	44.72***	15.19	58.98***	2.173***

Note:all variables express the sixteen crypto-currencies respective labels. LM statistic is used with respect tothe ARCH test, and the RS/mod statistic is used to detect long memory.***, **, * indicate the estimators' significance at the 1%, 5% and 10% levels, respectively.

Table 2
Goodness-of-fit estimation relevant to the different GARCH-family models.

	Augur			Bitshares			IOTA			Komodo			Monero		
Model	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL
FIEGARCH	−709.3	−682.0	361.7	−676.0	−648.7	345.0	−667.4	−640.1	340.7	−697.5	−670.2	355.7	−821.7	−794.4	417.9
FIGARCH	−712.2	−688.8	362.1	−672.7	−649.3	342.4	−664.5	−641.1	338.2	−701.6	−678.2	356.8	−794.4	−771.0	403.2
EGARCH-"ged"	−771.5	−744.2	392.7	−721.1	−693.8	367.6	−698.9	−671.6	356.5	−721.3	−694.0	367.7	−856.8	−829.5	435.4
TGARCH-"ged"	−773.4	−746.1	393.7	−722.5	−695.2	368.2	−697.6	−670.3	355.8	**−724.5**	**−697.2**	**369.2**	−858.9	−831.6	436.5
PGARCH-"ged"	−770.4	−743.1	392.2	−721.9	−694.6	368.0	−698.7	−671.4	356.3	−720.1	−692.8	367.1	−855.9	−828.6	435.0
EGARCH-"normal"	−715.7	−692.3	363.8	−681.0	−657.6	346.5	−668.7	−645.3	340.3	−698.7	−675.3	355.4	−793.4	−770.0	402.7
TGARCH-"normal"	−721.3	−697.9	366.7	−680.8	−657.4	346.4	−667.8	−644.4	339.9	−705.4	−682.0	358.7	−799.4	−776.0	405.7
PGARCH-"normal"	−715.2	−691.8	363.6	−682.2	−658.8	347.1	−668.7	−645.3	340.4	−697.6	−674.2	354.8	−794.3	−770.9	403.2
EGARCH-"student"	−772.3	−745.0	393.2	−713.6	−686.3	363.8	−696.0	−668.7	355.0	−721.7	−694.4	367.9	−859.8	−832.5	436.9
TGARCH-"student"	−773.0	−745.7	393.5	−715.7	−688.4	364.9	−694.8	−667.5	354.4	−724.2	−696.9	369.1	−859.7	−832.4	436.9
PGARCH-"student"	−771.4	−744.1	392.7	−714.1	−686.8	364.1	−695.6	−668.3	354.8	−720.8	−693.5	367.4	−858.0	−830.7	436.0
EGARCH-"double exp"	−773.3	−749.9	392.6	−723.1	−699.7	367.6	**−700.5**	**−677.1**	**356.3**	−719.8	−696.4	365.9	−858.8	−835.4	435.4
TGARCH-"double exp"	**−775.4**	**−752.0**	**393.7**	**−724.4**	**−701.0**	**368.2**	−699.3	−675.9	355.7	−722.2	−697.1	367.1	**−860.9**	**−837.5**	**436.9**
PGARCH-"double exp"	−772.5	−749.1	392.3	−723.9	−700.5	368.0	−700.2	−676.8	356.1	−718.8	−695.4	365.4	−857.9	−834.5	434.9

	NEO			QTUM			Ripple			Stratis			Waves		
Model	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL	AIC	BIC	LL
FIEGARCH	−636.9	−609.6	325.4	−631.3	−604.1	322.7	−811.9	−784.6	413.0	−646.2	−618.9	330.1	−766.1	−738.8	390.0
FIGARCH	−644.9	−621.5	328.4	−621.7	−598.3	316.9	−819.9	−796.5	415.9	−651.4	−628.0	331.7	−765.5	−742.1	388.8
EGARCH-"ged"	−707.2	−679.9	360.6	−703.8	−676.5	358.9	−870.3	−843.0	442.1	−669.6	−642.3	341.8	−809.8	−782.5	411.9
TGARCH-"ged"	−707.7	−680.4	360.6	−702.5	−675.2	358.3	−873.9	−846.6	443.8	**−671.0**	**−643.7**	**342.5**	−810.0	−782.7	412.0
PGARCH-"ged"	−706.9	−679.6	360.4	−704.0	−676.7	359.0	−869.2	−841.9	441.6	−669.8	−642.5	341.9	−809.2	−781.9	411.5
EGARCH-"normal"	−639.8	−616.4	325.9	−624.2	−600.8	318.1	−814.4	−791.0	413.2	−645.5	−622.1	328.8	−768.5	−745.1	390.3
TGARCH-"normal"	−642.5	−619.1	327.3	−625.0	−601.6	318.5	−822.5	−799.1	417.2	−649.3	−625.9	330.7	−766.8	−743.4	389.4
PGARCH-"normal"	−639.9	−616.5	325.9	−626.3	−602.9	319.1	−812.0	−788.6	412.0	−646.1	−622.7	329.1	−768.1	−744.7	390.1
EGARCH-"student"	−704.8	−677.5	359.4	**−704.7**	**−677.4**	**359.4**	−865.3	−838.0	439.7	−668.5	−641.2	341.3	−810.2	−782.9	412.1
TGARCH-"student"	−705.2	−677.9	359.6	−703.9	−676.6	359.0	−868.7	−841.4	441.4	−669.0	−641.7	341.5	−811.0	−783.7	412.5
PGARCH-"student"	−704.2	−676.9	359.1	−704.5	−677.3	359.3	−864.1	−836.8	439.0	−668.5	−641.2	341.2	−809.3	−782.0	411.5
EGARCH-"double exp"	−708.8	−685.4	360.4	−704.4	−671.0	358.2	−871.9	−848.5	442.0	−668.5	−642.1	340.3	−810.9	−787.5	411.4
TGARCH-"double exp"	**−709.5**	**−686.1**	**360.7**	−702.7	−677.3	357.3	**−875.8**	**−852.4**	**443.9**	−669.4	−643.0	340.7	**−811.2**	**−787.8**	**411.6**
PGARCH-"double exp"	−708.5	−685.1	360.3	−704.6	−671.2	358.3	−870.8	−847.4	441.4	−668.6	−643.2	340.3	−810.3	−786.9	411.2

Table 3
GARCH estimation models.

	Augur	Bitshares	Komodo	Monero	NEO	Ripple	Stratis	Waves	IOTA	QTUM
C	−0.0039	−0.0041	−0.0021	−0.0029	−0.0114***	−0.0061	−0.0055	−0.0061**	−0.0035	−0.0090**
AR(1)	−0.0947**	−0.0606	0.0135	−0.0749*	−0.0844**	−0.0219	−0.0293	−0.0462	−0.1005**	−0.0970**
A	0.0005*	0.0002	0.0004	0.0005	0.0002	0.0006	0.0004	0.0008	−0.5602*	−0.3763
ARCH(1)	0.1104**	0.0614*	0.1030**	0.1243*	0.0727*	0.1861	0.0998**	0.1730*	0.0712**	0.1572**
GARCH(1)	0.8539***	0.9178***	0.8803***	0.8364***	0.8817***	0.7401	0.8961***	0.7218***	0.9117***	0.9381***
GAMMA(1)	−0.0566***	−0.0893**	−0.0508***	−0.0810**	0.0493	−0.0309*	−0.0645*	0.0708	–	–
LEV(1)	–	–	–	–	–	–	–	–	0.3587	−0.0698*
JB	129.7***	78.03***	60.13***	1554***	271.6***	132.4***	29.65***	81.42***	67.92***	1698***
Q(12)	13.11	17.86	11.72	13.92	7.038	23.35	12.45	13.01	22.46	15.99
Q2(12)	19.21	16.64	13.57	3.156	2.738	10.7	23.81	7.516	5.457	14.83
LM(12)	0.0891	16.04	−0.2395	−1.649	−0.5196	12.36	23.1	6.904	5.073	15.05

Note:***, **, * indicate that the estimators are significant at the 1%, 5% and 10% levels, respectively.

autocorrelation could not be rejected. Finally, it is worth outlining that even though the different models' drawn residuals still depart from normality, the Jarque-Bera test associated value proves to be considerably significant.

9.2. The crypto-currencies' return volatilities: GARCH models

The results achieved following implementation of the asymmetric appropriate GARCH models figure on Table 3, below. In a first place, and concerning the asymmetric volatility parameter, the attained results reveal that most of the crypto-currencies tend to exhibit a negative leverage (GAMMA(1) and LEV(1)),except for the NEO, IOTA and Waves. The negative coefficients imply that negative shocks prove to increase volatility less than positive shocks do, which are in stark contrast with the stock markets' generally reported positive coefficients (Jeribi et al., 2015; Fakhfekh et al., 2016).In effect, the uninformed investors' trading activities appear to culminate in a noticeable rise in volatility, while the informed traders' activities prove to help in reducing it. In this regard, Avramov et al. (2006) argue that the prices' witnessed changes are mainly due to uninformed investors who contribute more remarkably in increasing volatility than the informed investors' related price changes would do. As a matter of fact, the inverted asymmetric effect, associated with the NEO, IOTA and Waves crypto-currencies, can have its explanation in the of uninformed investors related herding noticeable whenever prices tend to go up, and in the informed investors' inverse behavior whenever prices tend to go down. Similarly, the AR (1) related coefficients also indicate that the uninformed investors do play a significant role with regard to the entirety of the crypto-currencies under review, since the relevant AR (1) coefficients prove to be almost similar with respect to all the crypto-currencies, a finding that corroborates well with those documented by Avramov et al. (2006) as well as Baur and Dimpfl (2018).

Concerning the positive (inverted) asymmetric volatility effect, it holds noticeably for the NEO, IOTA and Waves crypto-currencies. This almost puzzling and unexpectedly strange result may well have its explanation in the fact that the largest crypto-currencies are not dominated by uninformed investors but, rather, by informed investors. Actually, the NEO, IOTA and Waves interested investors could be less prone to herding and act as contrarians, which might well explain the varying estimates, and there from behavior, of the two largest main crypto-currencies.

Hence, if uninformed investors drive up prices due to a missing out relating fear, associated with rising market prices, volatility will increase even more in falling markets. Similarly, if uninformed investors undertake to pump up prices, as part of a pump and dump scheme, volatility will certainly increase rather than fall in such markets. In such cases, uninformed investors are more inclined to drive up prices to levels that are reversed and corrected by informed investors, thus, giving rise to an asymmetric volatility effect. Asymmetry is also consistent with the disposition effect? in absence of informed investors, if uninformed investors turn out to be rather inclined to sell in rising market circumstances than in falling market conditions, implying intensified reversal and volatility to persist in rising markets, and non-reversal and lower volatility prevailing in falling markets.

As the informed traders are assumed to detain fundamental information concerning the asset's value, they will then coordinate their trading strategies accordingly. More particularly, they will undertake to buy the stock in undervalued circumstances, i. e., at a lower price, and resell it when the offered price exceeds the fundamental value. Such trading practices contribute noticeably in kindling price discovery, which would ultimately reveal the asset's actual fundamental value. Inversely, however, uninformed noise traders, also referred to as liquidity traders, trade for reasons other than exploiting information, such as the need to smooth consumption over time, thus, providing liquidity on the market. Actually, the informed and uninformed investors trading practices leave different traces on the return process of the traded asset. In this regard, Avramov et al. (2006) argue that the returns' serial correlation implies predominance of the uninformed investors' trading practices, whereas a zero autocorrelation implies well a prevalence of the informed investors' trading activities, and that the uninformed investors associated price changes will be more reversed by increasing volatility, rather than the informed investors' resultant price changes.

As compared to conventional assets, such as stocks and bonds, the Bitcoin prices are much more volatile (Balcilar et al., 2017) and particularly sensitive to regulatory and market events. In this respect, the relevant literature outlines that the Bitcoin returns do not seem to conform to the Efficient Market Hypothesis, and that they are not independent but predictable. This idea is further stressed by Urquhart (2017), who discovered the persistence of significant clustering evidence at round numbers, with over 10% of prices ending with 00 decimals compared to other variations, but no significant pattern of returns seems to prevail after the round number. He went even further by supporting the hypothesis of Harris (1991), highlighting that price and volume proves to display a significant positive relationship with price clustering at whole numbers.

10. Conclusion

Given the considerable interest oriented to the crypto-currencies, it is critically important to opt for the appropriate models fit for effectively estimating volatility. To this end, our focus of interest has been directed to the GARCH-family models enabling to help in efficiently accounting the long-memory and asymmetric effects.

Through applying fourteen GARCH specifications with different relating error distributions, namely, the FIGARCH, FIEGARCH, EGARCH, PGARCH, TGARCH models, we end up by reaching the result that the TGARCH with double exponential distribution turns out to be the most appropriate model fit for effectively describing the Augur, Bitshares, Monero, NEO, Ripple and Waves associated returns volatility series. As for the TGARCH model under generalized error distribution, it proves to stand as the most effective model, useful for efficiently expressing the Komodo and Stratis related return volatility series. Finally, the IOTA relevant returns are discovered to be rather effectively modulated by means of an EGARCH with double exponential distribution, while and the EGARCH under student distribution proves to represent the best approach fit for optimally conveying the QTUM return relating volatility.

In addition, it has also been discovered that volatility tends to increase as a response to positive shocks rather than to negative shocks, reflecting an asymmetric effect that differs noticeably from that often observed in stock markets. As a matter of fact, the increased volatility in response to positive shocks turns out to have its explanation in the uninformed investors related herding. Actually, our achieved findings may well be useful to investors intending to effectively account for potential volatility through implementing rather efficient hedging strategies.

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